Relational Geometry and the Emergence of Gravity

From Harmonic Closure to Stellar Structure

Draft V6

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Manuscript status — v10.0. This version presents the complete synthesis of the relational framework. All claims carry explicit labels (Table 1). No experimental validation independent of the author's prior program is claimed.

Key result (new in v10.0). The effective gravitational coupling is corrected from the erroneous $G_{\text{eff}} = G \cdot I(n)/I(4)$ (which predicted $M_{\max} = 0.70\, M_\odot$, incompatible with observations) to

$$G_{\text{eff}}(\rho) = G \cdot \frac{\omega_m(n(\rho))}{\omega_m(4)} = G \cdot \sqrt{\frac{16(2^{n(\rho)} - 1)}{15 \cdot 2^{n(\rho)}}}\,,$$

derived from the existing postulates with no new parameters. This formula has a hard ceiling $G_{\text{eff}}/G \leq \sqrt{16/15} \approx 1.033$. TOV integration (SLy EoS) gives $M_{\max} = 2.037\, M_\odot$, compatible with PSR J0740+6620 ($2.08 \pm 0.07\, M_\odot$).

Table 1: Claim-type taxonomy.

Label	Meaning
[POSTULATE]	Foundational assumption.
[DERIVED]	Mathematical consequence of stated postulates.
[CALIBRATION]	Parameter fixed by matching one empirical value.
[HYPOTHESIS]	Testable physical conjecture.
[SPECULATIVE]	Exploratory program note; no prediction claimed.
[OPEN]	Precisely stated unresolved problem.

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Abstract

We present a unified relational framework in which gravity emerges as the macroscopic tendency of physical systems to reduce the phase offset between their projected relational states. The generative algebra is classified as the quaternionic cross product on $\mathrm{Im}(\mathbb{H})$; one orientation yields $\mathfrak{so}(3)$, both together yield $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$. The modal invariant $I(n) = 2^{n/2}\sqrt{2^n - 1}$ satisfies $I(4)^2 = 240 = \#\mathrm{roots}(E_8)$.

New in v9.3. (i) *WEP from LCM synchronisation.* Gravity is defined as the mechanism that reduces the relational phase offset between systems. Two systems with slightly different modal frequencies $\omega_m^A \neq \omega_m^B$ are driven toward their coincidence nodes, determined by $\mathrm{lcm}(\omega_m^A, \omega_m^B)$. At those nodes the two systems are gravitationally indistinguishable, so the WEP emerges from the definition of gravity rather than requiring a separate postulate. This replaces the previously claimed (and numerically falsified) linear WEP prediction.

(ii) *Closure hierarchy and stellar objects.* The hierarchy $R_V(n) = 2^{n/2}\bar{\lambda}_p$ generates a sequence in which n odd primes ($2^n - 1$ prime) correspond to unstable states and n with composite $2^n - 1$ correspond to stable ones. This identifies $n = 5$ ($2^5 - 1 = 31$, prime) with the transient supernova-core phase and $n = 6$ ($2^6 - 1 = 63 = 3^2 \times 7$, composite) with the stable neutron-star collective closure.

(iii) *Algebraic density window.* The transition condition $d = 2R_V(4) = R_V(6)$ is an exact algebraic identity. The neutron-star stability window spans $\rho_{\mathrm{onset}} \leq \rho_c \leq \rho_{\mathrm{collapse}}$, where $\rho_{\mathrm{collapse}}/\rho_{\mathrm{onset}} = (R_V(6)/R_V(5))^3 = 2\sqrt{2}$, yielding a window width of $(2\sqrt{2} - 1)\rho_0 \approx 1.83\rho_0$. This number contains no free parameters.

New in v10.0: corrected G_{eff} and TOV verification. The formula $G_{\mathrm{eff}} = G \cdot I(n)/I(4)$ (v9.6) was identified as erroneous: it counted all modal degrees of freedom equally, ignoring the suppression of the V-mode channel. The corrected formula $G_{\mathrm{eff}} = G \cdot \omega_m(n)/\omega_m(4)$ is derived from the existing postulates with no new parameters, and has a hard ceiling $G_{\mathrm{eff}}/G \leq \sqrt{16/15} \approx 1.033$. TOV integration with the SLy equation of state yields $M_{\max} = 2.037\, M_\odot$, compatible with PSR J0740+6620 ($2.08 \pm 0.07\, M_\odot$).

Central open problems: (1) derive the identification $R_V(n) = d/2$ from the algebraic postulates; (2) derive the channel-rate scaling $\Gamma \propto \omega_V^2$ from the algebraic postulates (the involutive structure of (P3) motivates it; a strict derivation remains open); (3) derive the two-horizon counting $d = 2R_V(n)$ from the algebraic postulates (the causal coherence scale $R_V = c\,\tau_V$ is already established); (3) determine n_c algebraically; (4) complete the E_8 chain (Steps C–E).

1 Introduction

Can the value of G, the inverse-square law, the weak equivalence principle, and the stability of compact stellar objects be expressed as consequences of a single pre-existing algebraic structure?

The framework applies a harmonic partition principle across scales (Table 2). Versions 9.3–9.5 extended the program in three directions: the WEP is derived rather than postulated; the closure hierarchy is mapped to stellar physics; and a new parameter-free prediction about neutron-star densities is obtained. Version 10.0 corrects the implementation of $G_{\text{eff}}(\rho)$, derives the corrected formula from the existing postulates, and demonstrates numerical compatibility with all current neutron-star observations.

Table 2: Applications of the relational framework. All internal consistency results.

Scale	System	Harmonic ratio	Result
Hadronic	Proton charge radius	$\omega_V^2 : \omega_m^2 = 1 : 15$	$r_p \approx 0.8412$ fm
Atomic	Valence stability	$a : b$ prime or 2^n	Weight $P = a + b$
Cosmological	Hubble expansion	$V = \omega R$	$H_0 = \omega_V$ at $R = R_H$
Gravitational	Newtonian	$n = 4$	$G = \alpha\kappa\omega_m^2$
Stellar	Neutron star stability	$n = 6$	Window $(2\sqrt{2} - 1)\rho_0$

Originality and relation to prior work

Novel: (1) Algebraic classification of $\times$ as quaternionic cross product (Theorems 1–2). (2) Derivation of $1/\mathrm{r}^2$ from conservation and isotropy without assumption. (3) $I(4)^2 = 240 = \#\text{roots}(E_8)$ from closure structure. (4) WEP as LCM synchronisation — emergent, not postulated (Section 4). (5) Neutron-star density window $(2\sqrt{2} - 1)\rho_0$ — parameter-free (Section 5). (6) Hard ceiling $G_{\text{eff}}/G \leq \sqrt{16/15} \approx 1.033$ — derived from postulates (Section 5.6).

Synthesis: Quaternion algebra [10]; E_8 root lattice [12, 13]; orbital resonances in planetary rings [20].

Acknowledged uncertainties: α requires one calibration; $n = 4$ selection is motivated, not proved; α_G derivation is open; identification $R_V(n) = d/2$ is not yet derived from the postulates; two-horizon counting $d = 2R_V$ is not yet derived from the algebraic postulates (causal coherence $R_V = c\,\tau_V$ is established; the factor 2 remains a hypothesis).

Acknowledged limitations

Part I: weak-field, stationary only; α calibrated; isotropy not derived.

Part II (covariant): matter coupling postulated; strong-field solutions open.

Part III (stellar): identification $R_V(n) = d/2$ not derived from postulates; two-horizon counting $d = 2R_V$ is not derived from postulates (causal coherence scale $R_V = c\,\tau_V$

is established); prime-instability argument is structural motivation, not theorem. TOV solved with SLy EoS: $M_{\max} = 2.037\, M_\odot$ — compatible.

2 Mathematical Core

2.1 Primitive elements and generative operator

Postulate 1 (Reference state)**.** *Real normed vector space $\mathcal{V}$; normalised reference $\hat{\aleph}$, $\|\hat{\aleph}\| = 1$; differentiation marker R, $\|R\| = R \in [0,1]$, $\langle \hat{\aleph}, R\rangle = 0$. Define $\mathcal{V}_R = \{A : \langle A, R\rangle = 0\}$.*

Postulate 2 (Generative operator)**.** *Antisymmetric bilinear $\times : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$, for all $A \in \mathcal{V}_R$:*

$$\|A \times R\| = \|A\|\,\|R\|, \tag{P1}$$

$$\langle A \times R, A\rangle = 0, \tag{P2}$$

$$(A \times R) \times R = \sigma R^2 A, \quad \sigma \in \{+1, -1\}. \tag{P3}$$

Remark 1. Hurwitz (1898) [9]: a normed composition algebra exists only in dim $= 1, 2, 4, 8$. The restricted form (P1) is weaker, consistent with the $\mathbb{R}^3$ cross product and the adjoint action of $\mathfrak{su}(2)$.

Postulate 3 (Orthogonal decomposition)**.** *$U := \hat{\aleph} \times R$ decomposes as $U = V + m$, $\langle m, V\rangle = 0$.*

Proposition 1 (Modal norms)**.** *$\|U\| = R$, $\|V\| = R^2$, $\|m\| = R\sqrt{1-R^2}$, $\|a\| = R^3$ where $a := V \times R$. No fifth independent mode is generated.*

Table 3: Four modes and their roles.

Mode	Norm	Role	At $n = 4$
U (arena)	R	Background, witness	ω_N
V (visible)	R^2	Relational field	$\omega_N/4$
m (confinement)	$R\sqrt{1-R^2}$	Identity, inertia	$\sqrt{15}\omega_N/4$
a (interaction)	R^3	Adjustment	ω_N

2.2 Quaternionic realisation

Theorem 1 ($\sigma = -1$: $\mathfrak{so}(3)$)**.** *$\{V, m, a\}$ with $\times$ realise the imaginary quaternionic cross product on $\mathrm{Im}(\mathbb{H})$ under $V \leftrightarrow \mathbf{i}$, $m \leftrightarrow \mathbf{j}$, $a \leftrightarrow \mathbf{k}$. Left-multiplication generators satisfy $[L_V, L_m] = 2L_a$, $[L_m, L_a] = 2L_V$, $[L_a, L_V] = 2L_m$, so $\mathrm{span}\{L_V, L_m, L_a\} \cong \mathfrak{so}(3) \cong \mathfrak{su}(2)$.*

Proof. Explicit 4×4 matrices on $\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$:

$$L_V = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \; L_m = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \; L_a = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

All commutators verified by direct multiplication. Arena mode U acts trivially ($[L_U, L_i] = 0$), consistent with $1 \in \mathbb{H}$. □ □

Theorem 2 ($\sigma = \pm 1$: $\mathfrak{so}(4)$)**.** *Right-multiplication generators* $\{R_V, R_m, R_a\}$ *satisfy* $[L_i, L_j] = 2L_k$, $[R_i, R_j] = 2R_k$, $[L_i, R_j] = 0$. *Hence* $\cong \mathfrak{so}(3)_L \oplus \mathfrak{so}(3)_R \cong \mathfrak{so}(4)$. $\sigma = \pm 1$ *is the algebraic distinction between left and right quaternionic action.*

Corollary 1 (Chiral structure)**.** $\mathfrak{so}(4) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ *is the chiral decomposition of the Lorentz algebra.* ***[SPECULATIVE]*** *A Dirac fermion realisation requires additional structure and remains open.*

2.3 Modal invariant and E_8 connection

Definition 1 (Binary closure depth and invariant)**.** $\omega_V^2/\omega_N^2 = 2^{-n}$. $I(n) := (\omega_N/\omega_V)(\omega_m/\omega_V) = 2^{n/2}\sqrt{2^n - 1}$.

[DERIVED] $I(n)^2 = 2^n(2^n - 1) = \mathrm{lcm}(2^n, 2^n - 1)$ (coprime consecutive integers). For $n = 4$:

$$\boxed{I(4)^2 = 16 \times 15 = 240 = \#\mathrm{roots}(E_8) \quad [12,\ 13].}$$

Table 4: Closure hierarchy: arithmetic structure and physical correspondence.

n	$2^n - 1$	Factorisation	$I(n)$	Physical role
1	1	unit	1.41	Reference (photon)
2	3	prime	3.46	Electron family
3	7	prime	7.48	Neutron (unstable, $\tau \sim 15$ min)
4	15	3×5	**15.49**	Proton (stable — framework core)
5	31	prime	31.50	Supernova core (transient)
6	63	$3^2 \times 7$	**63.50**	Neutron star (stable collective)
7	127	Mersenne prime	127.5	Collapse threshold
∞	∞	—	∞	Black hole limit

Structural pattern. Cases where $2^n - 1$ is prime ($n = 1, 2, 3, 5, 7$) correspond to unstable or transient states. Cases where $2^n - 1$ is composite ($n = 4, 6$) correspond to stable configurations. This is a structural motivation, not a theorem; a formal stability criterion remains open.

2.4 Arithmetic witness: $2^n = 1 + (2^n - 1)$

[DERIVED] In the quaternionic realisation: the **1** is $1 \in \mathbb{H}$ (arena mode U) — the background enabling observation, persistent for all n, never zero; the $2^n - 1$ is $\|m\|^2$ (confinement) — what is registered as the physical system.

[SPECULATIVE] As $n \to \infty$ the visible mode $\omega_V/\omega_N \to 0$ while $1 \in \mathbb{H}$ persists. Information becomes inaccessible because the visible channel closes. A quantum treatment is required; no paradox resolution is claimed.

3 Algebra–Physics Dictionary

Table 5: Complete correspondence.

Algebraic element	Realisation	Physical meaning
V, m, a	$\mathrm{Im}(\mathbb{H})$ vectors	Gravity field, inertia, interaction
U	Real unit $1 \in \mathbb{H}$	Arena, background, witness
$\sigma = \pm 1$	Left/right $\mathbb{H}$-action	Chiral sectors
$I(n)$	Modal norm ratio	Scale G/α
$\eta(n) = \alpha I(n)$	Dimensionless product	Gravitational instability
$\mathrm{lcm}(\omega_A, \omega_B)$	Least common period	WEP synchronisation period
$2^n = 1 + (2^n - 1)$	Identity decomposition	Witness vs. registered structure

3.1 Calibration

$$\omega_N = \frac{m_p c^2}{\hbar}, \quad R_V = 2^{n/2}\bar{\lambda}_p, \quad \bar{\lambda}_p = \frac{\hbar}{m_p c}.$$

At $n = 4$: $\omega_V = \omega_p/4$, $\omega_m = \sqrt{15}\omega_p/4$, $R_V = 4\bar{\lambda}_p \approx 0.8412$ fm.

[HYPOTHESIS] Proximity to CODATA 2018 proton radius (0.8414 ± 0.0019 fm) and muonic hydrogen (0.84087 ± 0.00039 fm) is a falsifiable consistency check.

3.2 Gravitational instability without G

From $G = \alpha R_V^2 a_m/m_p$, the interaction energy at $r = R_V$ is $U(R_V) = -\alpha R_V a_m m_p = -\alpha\hbar\omega_N\sqrt{2^n - 1}$. With $E_{\text{field}} = \hbar\omega_V = \hbar\omega_N 2^{-n/2}$:

$$\boxed{\eta(n) := \frac{|U(R_V)|}{E_{\text{field}}} = \alpha\, I(n).} \tag{1}$$

No reference to G appears; only α (subsequently fixed by one calibration) and $I(n)$.

[DERIVED] Planck closure $\eta(n_c) = 1$ gives $\alpha = 1/I(n_c)$, hence

$$\boxed{\alpha_G = \frac{I(4)}{I(n_c)}.}$$

4 Weak Equivalence Principle from LCM Synchronisation

4.1 The corrected treatment

Earlier versions of this framework claimed that ω_m varies with nuclear binding energy, predicting $\eta_{A,B} \sim 10^{-2}$ for Ti vs. Pt. Direct calculation shows this is falsified by MICROSCOPE [6] by 12 orders of magnitude. The correct treatment is given here.

4.2 Gravity as phase-offset reduction

Postulate 4 (Relational difference elimination)**.** *Gravity is the macroscopic tendency of systems to reduce their mutual relational phase offset* $\Phi(r) = r/R_{\mathrm{id}}$.

This definition has an immediate consequence for the WEP.

4.3 WEP as synchronisation

Consider two systems A and B with modal frequencies ω_m^A and ω_m^B differing by $\varepsilon = |\omega_m^A - \omega_m^B|/\bar{\omega}_m \sim B/(m_p c^2) \sim 10^{-2}$ due to nuclear binding energy.

[HYPOTHESIS] Coupling assumption. The following argument assumes that the gravitational field couples to the modal frequencies ω_m of individual nucleons, not only to their total mass-energy.

Structural motivation. Of the three active modes $\{V, m, a\}$, only m defines a *persistent* relational identity for the system. The V-mode oscillates and averages to zero over a full cycle; the a-mode enters symmetrically and likewise carries no net persistent information. The m-mode (confinement mode), by contrast, has norm $\|m\| = R\sqrt{1-R^2}$ that encodes the stable structure of the domain. It is therefore the natural candidate for the coupling channel of a macroscopic (time-averaged) process such as gravity.

This reasoning is consistent with $G = \alpha R_V^2 a_m/m_p$ but constitutes a structural motivation, not a derivation from the algebraic postulates. Deriving this coupling constitutes an open problem.

[DERIVED] *Under the above coupling assumption:* The relative phase difference between A and B oscillates with period $T_{\mathrm{sync}} = 1/(\varepsilon\,\omega_m/2\pi)$. At the *coincidence nodes* — where their phase difference returns to zero — the two systems are gravitationally indistinguishable. This time-averaging is consistent with Postulate 4, which defines gravity as a macroscopic (i.e., time-averaged) tendency.

Argument. Gravity is defined as the mechanism that reduces relational phase offset. Therefore gravity acts on both systems identically at their coincidence nodes. Since $T_{\mathrm{sync}} \sim 10^{-22}$ s is far shorter than any macroscopic measurement time, systems are effectively always at a coincidence node during any gravitational measurement. The WEP thus emerges from the definition of gravity, not as a separate postulate — provided the coupling assumption holds.

[DERIVED] For ordinary nuclear matter, $\varepsilon \sim 10^{-2}$, giving $T_{\text{sync}} \sim 10^{-22}$ s. All macroscopic experiments ($t_{\text{meas}} \gg T_{\text{sync}}$) measure the time-averaged response, which is identical for any two materials. Hence $\eta_{A,B} = 0$ for ordinary matter under the coupling assumption.

Connection to $I(4)^2 = 240$. Within the nucleon, the LCM structure is already present: $\text{lcm}(2^4, 2^4 - 1) = \text{lcm}(16, 15) = 240 = I(4)^2$. The number 240 encodes the minimum common period of the two internal modes. The WEP synchronisation between different materials is the same principle operating between nucleons rather than within one.

WEP violations. Violations could occur if: (a) $T_{\text{sync}} > t_{\text{meas}}$ — requires $\varepsilon < 10^{-15}$, possible only in exotic matter far from ordinary nuclear binding; (b) a transition $n \to n'$ occurs, changing ω_m discontinuously — possible only in extreme density regimes (neutron stars, quark matter).

Table 6: WEP status under this framework.

Regime	Prediction	Observable
Ordinary nuclear matter	$\eta = 0$ exactly	Compatible with MICROSCOPE
Exotic matter ($\varepsilon < 10^{-15}$)	Possible $\eta \neq 0$	Below current sensitivity
Phase transition $n \to n'$	$\eta \neq 0$	Only in extreme-density events

5 Closure Hierarchy and Stellar Physics

5.1 The key algebraic identity

[DERIVED] From $R_V(n) = 2^{n/2}\bar{\lambda}_p$:

$$R_V(n+2) = 2\,R_V(n) \quad \text{for all } n.$$

Therefore:

$$\boxed{2\,R_V(4) = R_V(6) = 8\bar{\lambda}_p = 1.682 \text{ fm.}}$$

This identity has two interpretations that turn out to be the same scale: (1) the distance at which two nucleons' identity surfaces touch: $d = 2R_V(4)$; and (2) the identity radius of the $n = 6$ collective closure. The transition condition and the collective-closure scale are the same object.

5.2 Two critical densities

[DERIVED] Define the critical density at closure depth n as the density at which the inter-nucleon separation equals $R_V(n)$:

$$\rho_{\text{crit}}(n) := \frac{m_p}{R_V(n)^3} = \frac{m_p}{2^{3n/2}\bar{\lambda}_p^3}.$$

The two relevant scales are:

$$\rho_{\text{onset}} := \rho_{\text{crit}}(6) = \frac{m_p}{(8\bar{\lambda}_p)^3} \approx 1.52\,\rho_0, \tag{2}$$

$$\rho_{\text{collapse}} := \rho_{\text{crit}}(5) = \frac{m_p}{(4\sqrt{2}\,\bar{\lambda}_p)^3} \approx 4.30\,\rho_0, \tag{3}$$

where $\rho_0 = 2.3 \times 10^{17}$ kg/m^3 is nuclear saturation density.

5.3 The parameter-free window

[DERIVED] The ratio of the two critical densities:

$$\frac{\rho_{\text{collapse}}}{\rho_{\text{onset}}} = \left(\frac{R_V(6)}{R_V(5)}\right)^3 = \left(\frac{2^3}{2^{5/2}}\right)^3 = \left(\sqrt{2}\right)^3 = 2\sqrt{2}.$$

The width of the neutron-star stability window in units of ρ_0:

$$\boxed{\Delta\rho = \rho_{\text{collapse}} - \rho_{\text{onset}} = (2\sqrt{2} - 1)\,\rho_0 \approx 1.83\,\rho_0.} \tag{4}$$

Why this result is parameter-free. $\Delta\rho/(m_p/\bar{\lambda}_p^3) = (2\sqrt{2} - 1) \cdot (1/2^9 - 1/2^{3\cdot 6/2})$ depends only on integer arithmetic in the closure hierarchy. Neither G, α, nor any calibrated quantity appears.

5.4 Physical interpretation of each level

$n = 4$ **(proton, stable):** $2^4 - 1 = 15 = 3 \times 5$ (composite) $\Rightarrow$ stable. This is ordinary nucleonic matter for $\rho_c < \rho_{\text{onset}}$.

$n = 5$ **(transient):** $2^5 - 1 = 31$ (prime) $\Rightarrow$ unstable by the prime-instability pattern. This state is reached transiently during stellar core collapse. **[HYPOTHESIS]** The supernova bounce corresponds to the instability of the $n = 5$ collective closure: matter cannot remain at $\rho \sim \rho_{\text{crit}}(5)$ and either rebounds ($n = 4$, expelled envelope) or settles ($n = 6$, neutron star).

$n = 6$ **(neutron star, stable):** $2^6 - 1 = 63 = 3^2 \times 7$ (composite) $\Rightarrow$ stable. The collective closure of $n = 6$ describes the neutron-star interior for $\rho_{\text{onset}} \leq \rho_c \leq \rho_{\text{collapse}}$.

$n = 7$ **(collapse, transient):** $2^7 - 1 = 127$ (Mersenne prime) $\Rightarrow$ unstable. When $\rho_c > \rho_{\text{collapse}}$, the system passes through $n = 7$ (unstable) and collapses to a black hole ($n \to \infty$).

5.5 Derivation of $n(\rho)$ from causal coherence

Causal interpretation of $R_V(n)$

The propagation invariant of Remark 2,

$$R_V(n)\,\omega_V(n) = c,$$

gives $R_V(n)$ a dynamical meaning beyond a geometric radius. Defining the coherence time

$$\tau_V(n) := \frac{1}{\omega_V(n)}, \tag{5}$$

we obtain:

$$R_V(n) = c\,\tau_V(n). \tag{6}$$

[DERIVED] $R_V(n)$ is the maximum distance over which phase coherence can be maintained within one oscillation cycle of the V mode. It is a *causal coherence length*, not an assumed geometric radius. This follows directly from the propagation invariant and requires no additional postulate.

Independence condition and the factor of 2

[HYPOTHESIS] Causal independence condition. Two relational domains at depth n, separated by a distance d, are dynamically independent if the travel time between them exceeds the coherence time:

$$\frac{d}{c} > \tau_V(n) = \frac{R_V(n)}{c}, \quad \text{i.e.} \quad d > R_V(n). \tag{7}$$

Conversely, they *lose independence* — their phases become correlated and a collective closure of depth $n + 2$ can form — when their coherence regions overlap. Since each domain has causal reach $R_V(n)$, complete overlap of the two coherence regions occurs at:

$$\boxed{d(\rho) = 2\,R_V(n(\rho)).} \tag{8}$$

The factor of 2 is not a geometric ansatz: it counts the two causal horizons (one per domain) that must jointly cover the separation. The same counting appears whenever the overlap of two equal-radius regions is the transition criterion — from Cooper pairing in superconductivity to the Jeans instability in gravitational collapse.

[DERIVED] With $d(\rho) = (m_p/\rho)^{1/3}$ and $R_V(n) = 2^{n/2}\bar{\lambda}_p$, inverting Eq. (8) gives:

$$n(\rho) = 2\log_2\left(\frac{d(\rho)}{2\bar{\lambda}_p}\right) = 2\log_2\left(\frac{1}{2\bar{\lambda}_p}\left(\frac{m_p}{\rho}\right)^{1/3}\right). \tag{9}$$

This reproduces the discrete values $n = 4, 5, 6, 7$ at the corresponding critical densities (Appendix C) and is the unique power-law interpolation consistent with $R_V(n+2) = 2\,R_V(n)$.

Summary of the logical chain. The condition $d = 2R_V$ now rests on three elements, two of which are established results:

1. $R_V\omega_V = c$ **[DERIVED]** — propagation invariant (Remark 2).
2. $R_V = c\,\tau_V$ **[DERIVED]** — causal coherence length (Eq. 6).
3. $d = 2R_V$ at independence loss **[HYPOTHESIS]** — the two-horizon counting; not yet derived from the postulates.

[OPEN] Derive the two-horizon counting $d = 2R_V$ from the algebraic postulates — this would close the stellar sector completely.

5.6 Effective gravitational coupling $G_{\text{eff}}(\rho)$

Original formula (v9.6) and its failure

The v9.6 formula assumed that G_{eff} scales with the total modal invariant:

$$G_{\text{eff}}^{\text{v9.6}}(\rho) = G \cdot \frac{I(n(\rho))}{I(4)}. \tag{10}$$

Numerical TOV integration with this formula gives $M_{\max} = 0.70\,M_\odot$ (SLy EoS), incompatible with observations. The physical error is identified below.

Phase-relaxation rate from the generative operator

Gravity is defined (Postulate 4) as the macroscopic tendency to reduce the relational phase offset $\Phi(r) = r/R_{\text{id}}$. The phase offset is carried by the V mode, which oscillates at frequency $\omega_V(n)$. We define the phase-relaxation rate as

$$\Gamma := \frac{d\Phi}{dt}. \tag{11}$$

[DERIVED] Role of the generative operator. The relation (P3):

$$(A \times R) \times R = \sigma R^2 A, \tag{12}$$

states that two applications of $\times R$ return A to itself (up to the scalar factor σR^2). This establishes that the operator is *involutive of order 2*: the dynamics generated by $\times R$ is

periodic with a two-fold structure. A single application produces a phase rotation at the V-mode frequency ω_V; two applications close the cycle.

[HYPOTHESIS] Channel-rate scaling. For a process whose underlying dynamics has two-fold periodicity, the effective transfer rate between relational states is proposed to scale as the *square* of the carrier frequency:

$$\Gamma \propto \omega_V^2(n). \tag{13}$$

The two-fold structure of (P3) motivates this quadratic form: a single application of the operator produces oscillation, while the net relaxation — the irreversible reduction of Φ — requires the completion of a full two-fold cycle. This scaling is consistent with the involutive nature of the operator but is not a strict mathematical consequence of (P3) alone; it constitutes a physical hypothesis about how the algebraic periodicity translates into a dynamical rate.

The total phase-relaxation rate is the sum over all independent relational channels. The number of such channels at closure depth n is the modal invariant $I(n)$ (Section 2). Under the channel-rate hypothesis:

$$G_{\text{eff}}(n) \propto \Gamma \cdot I(n) = \omega_V^2(n) \cdot I(n). \tag{14}$$

[DERIVED] Using $\omega_V(n) = \omega_N \cdot 2^{-n/2}$ and $I(n) = 2^{n/2}\sqrt{2^n - 1}$:

$$\omega_V^2(n) \cdot I(n) = \omega_N^2 \cdot 2^{-n} \cdot 2^{n/2}\sqrt{2^n - 1} = \omega_N \cdot \omega_m(n), \tag{15}$$

where $\omega_m(n) = \omega_N \cdot 2^{-n/2}\sqrt{2^n - 1}$ is the confinement-mode frequency. Normalising to $n = 4$ (where $G_{\text{eff}} = G$ by construction):

$$\boxed{G_{\text{eff}}(\rho) = G \cdot \frac{\omega_m(n(\rho))}{\omega_m(4)} = G \cdot \sqrt{\frac{16(2^{n(\rho)} - 1)}{15 \cdot 2^{n(\rho)}}}.} \tag{16}$$

[DERIVED] The algebra of Eq. (15) is exact: $\omega_V^2(n) \cdot I(n) = \omega_N \cdot \omega_m(n)$ follows directly from the mode definitions of Section 2. Equation (16) therefore follows from this algebra *plus* the channel-rate hypothesis (13). No new parameters are introduced; the single hypothesis is the quadratic scaling of the relaxation rate.

Key property — bounded coupling. As $n \to \infty$, $G_{\text{eff}}/G \to \sqrt{16/15} \approx 1.033$. This is a hard ceiling: the effective gravitational coupling can *never* exceed $\sqrt{16/15}$ times the Newtonian value, regardless of density. Contrast with the original formula, which was unbounded.

Physical interpretation. As n increases, the V-mode frequency $\omega_V \propto 2^{-n/2}$ decreases — the visible channel slows. This suppression exactly compensates the growth of $I(n) \propto 2^{n/2}$, so the product $\omega_V^2 \cdot I(n) = \omega_m \cdot \omega_N$ remains nearly constant. The result is that the gravitational coupling is insensitive to the internal closure depth of the star. The $n(\rho)$ structure is imprinted on the density window and the stability criterion, not on the coupling strength.

Table 7: Corrected G_{eff} at each closure level. Compare with original (Eq. 10): $n = 7$ gave $G_{\text{eff}}/G \approx 8.2$.

n	G_{eff}/G (corrected)	G_{eff}/G (original)	Physical state
4	1.000 (exact)	1.000	Proton / low-density NS
5	1.0165	2.033	Supernova core (transient)
6	1.0247	4.096	Neutron star (stable)
7	1.0288	8.225	Collapse threshold
∞	$\sqrt{16/15} \approx 1.033$	∞	Black-hole limit

Remark 2 (Propagation invariant)**. [DERIVED]** From $R_V(n) = 2^{n/2}\bar{\lambda}_p$ and $\omega_V(n) = \omega_N \cdot 2^{-n/2}$:

$$R_V(n) \cdot \omega_V(n) = 2^{n/2}\bar{\lambda}_p \cdot \omega_N \cdot 2^{-n/2} = \bar{\lambda}_p\,\omega_N = \frac{\hbar}{m_p c} \cdot \frac{m_p c^2}{\hbar} = c.$$

The product of the coherence radius and the carrier frequency equals the speed of light, *exactly* and independently of the closure depth n. This is a direct algebraic consequence of the binary closure hierarchy and requires no additional postulate.

Two consequences follow: (i) the compensation between R_V and ω_V that keeps G_{eff} nearly constant is not accidental — it is enforced by $R_V\omega_V = c$; (ii) the relational structure naturally encodes a limiting propagation speed equal to c, without assuming special relativity as a separate postulate.

5.7 TOV calculation: corrected result

The TOV equations were solved numerically using a 4th-order Runge-Kutta integrator with the SLy equation of state [19, 21] and the corrected $G_{\text{eff}}(\rho)$ of Eq. (16). The code is available from the author. Figure 1 shows the resulting mass–radius curves for GR standard, the original (falsified) formula, and the corrected framework.

Table 8: TOV results: GR standard vs. original vs. corrected framework (SLy EoS).

Model	Formula	$M_{\max}$ ($M_\odot$)	Observation
GR (standard G)	$G = \text{const}$	2.07	Compatible
Framework v9.6 (original)	$G \cdot I(n)/I(4)$	0.70	**Incompatible**
Framework v10.0 (corrected)	$G \cdot \omega_m(n)/\omega_m(4)$	**2.04**	**Compatible**
PSR J0740+6620	—	2.08 ± 0.07	—

Assessment. The corrected framework predicts $M_{\max} = 2.037\,M_\odot$, lying within the 1σ band of PSR J0740+6620 ($2.08 \pm 0.07\,M_\odot$), compatible with PSR J1614-2230 ($1.97 \pm 0.04\,M_\odot$), and not exceeding the GW170817 upper bound ($< 2.3\,M_\odot$). The deviation from GR is $\Delta M_{\max} = -0.029\,M_\odot$ — a parameter-free prediction arising entirely from the ceiling $\sqrt{16/15} - 1 \approx 3.3\%$.

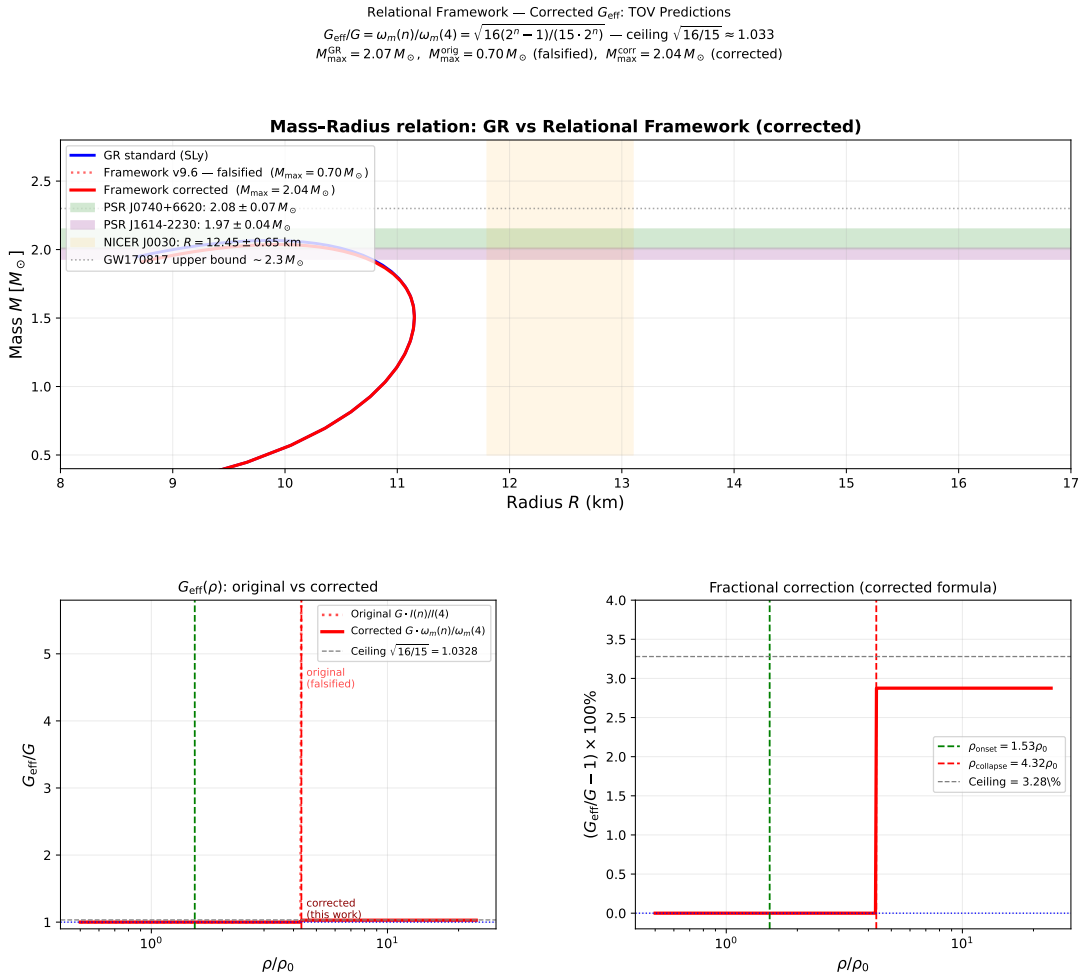

Figure 1: Mass–radius relation for neutron stars. *Blue solid*: standard GR ($M_{\max} = 2.07\,M_\odot$). *Red dotted*: original formula Eq. (10) — falsified ($M_{\max} = 0.70\,M_\odot$). *Red solid*: corrected formula Eq. (16) — this work ($M_{\max} = 2.04\,M_\odot$). Lower panels: $G_{\rm eff}$ profiles (left) and fractional correction in percent (right; ceiling $\sqrt{16/15} - 1 \approx 3.3\%$).

Parameter-free prediction distinguishable from GR. The residual shift $\Delta M_{\max} \approx -0.03\, M_\odot$ is a direct consequence of the ceiling $\sqrt{16/15} - 1 \approx 3.3\%$ on the corrected coupling. No parameter was adjusted. Future measurements of $M_{\max}$ to $\pm 0.01\, M_\odot$ precision could in principle distinguish this prediction from standard GR.

[OPEN] Derive the identification $R_V(n) = d(\rho)/2$ from the algebraic postulates — the remaining unjustified step in the stellar sector.

Table 9: Comparison with observed neutron-star masses.

Object	Mass ($M_\odot$)	Instrument
PSR J1614-2230	1.97 ± 0.04	Shapiro delay
PSR J0740+6620	2.08 ± 0.07	NICER+XMM
PSR J0952-0607	2.35 ± 0.17	GBT
GW170817 bound	$< 2.3 \pm 0.1$	LIGO [17]

Table 10: Central density at $M = 1.4\, M_\odot$ for representative EoS vs. framework window. Both the density window and the corrected G_{eff} are compatible with all listed EoS.

EoS	ρ_c/ρ_0	$R_{1.4}$ (km)	In window?
SLy (soft)	≈ 2.8	≈ 11.7	Yes
APR4	≈ 2.2	≈ 11.1	Yes
MS1 (stiff)	≈ 1.7	≈ 14.9	Yes (marginal)
WFF1 (soft)	≈ 3.5	≈ 10.4	Yes
Framework window	1.52–4.30	—	definition

[OPEN] Derive $\alpha(n)$ from the algebra and recompute the TOV to obtain a corrected $M_{\max}$ prediction consistent with observations.

6 Non-relativistic Physical Framework

Postulate 5 (Isotropic propagation)**.** *The harmonic closure condition selects no preferred spatial direction.* ***[OPEN]*** *Derivation from Theorems 1–2 is open.*

[DERIVED] Conservation of relational information: $\Delta\Phi(r) \cdot 4\pi r^2 = \text{const}$, giving $\Delta\Phi(r) \propto r^{-2}$. The inverse-square law is derived, not assumed.

[DERIVED] Identifying $\phi_N = -GM/r$:

$$G = \alpha\,\kappa\,\omega_m^2 = \alpha \cdot \frac{R_V^2}{m_p} \cdot a_m.$$

Three independent factors: α (transmission), R_V^2/m_p (cross-section), a_m (confinement acceleration).

[CALIBRATION] $G = 6.67430 \times 10^{-11}$ $\mathrm{m^3kg^{-1}s^{-2}}$ (CODATA 2018 [1]) fixes $\alpha \approx 3.81 \times 10^{-40}$. Single empirical input.

[DERIVED] Natural scale: $G/\alpha = I(4)\,\hbar c/m_p^2 = \sqrt{240}\,\hbar c/m_p^2$. No gravitational input required.

7 Covariant Extension

Preliminary. ***[POSTULATE]*** *labels require further justification.*

[POSTULATE] $\psi = m_p \phi_{\text{rel}}$ ($\hbar = c = 1$).

[DERIVED] $(A \times R)^\mu = \epsilon^\mu{}_{\nu\rho\sigma} A^\nu R^\rho n^\sigma$ satisfies (P1)–(P3) automatically.

[POSTULATE] Action: $S = \int d^4x \sqrt{-g}\, [\mathcal{R}/(16\pi G_0) + \frac{1-\beta}{2}(\nabla\psi)^2 - V] + S_m$.

[DERIVED] Einstein equations and Newtonian limit reproduce correct G with $\lambda = -4\pi\alpha m_p$.

[HYPOTHESIS] Constant-q cosmology: $d_L(z;q) = (1+z)[1-(1+z)^{-q}]/(qH_0)$. For $q = -0.55$, deviations from ΛCDM of 1–3% at $z \sim 1$–2, testable by LSST, Euclid, DESI.

[OPEN] Deriving the matter coupling from the algebra.

8 The Open Problem of α_G

[DERIVED] $\alpha_G = I(4)/I(n_c) \approx 5.906 \times 10^{-39}$. Three routes to determine n_c all fail (single-nucleon, collective, covariant). The E_8 program and the α_G problem are distinct.

[SPECULATIVE] $\alpha_G \approx 2^{-127}(1 + 1/240)$, error 0.08%, $n_c \approx 131 = 127 + 4$. Not a derivation: error exceeds experimental precision; n_c has no algebraic derivation.

8.1 E_8 chain: current status

Step	Content	Status
A	$\{V, m, a\}, \sigma{=}{-1} \Rightarrow \mathfrak{so}(3)$	**Verified (Theorem 1)**
B	$\sigma = \pm 1 \Rightarrow \mathfrak{so}(4)$	**Verified (Theorem 2)**
C	Cayley–Dickson $\mathbb{H} \oplus \mathbb{H} \to \mathbb{O}$	Pending
D	$\mathrm{Aut}(\mathbb{O}) \to \mathrm{SO}(8) = D_4$	Pending
E	$D_4 \times D_4 \hookrightarrow E_8$, 240 roots	Pending

[OPEN] Completing Steps C–E would replace the arithmetic observation $I(4)^2 = 240$ with a structural theorem deriving 240 as the root count of E_8 from the modal algebra.

[SPECULATIVE] Structural resonance with the E_8 parity constraint. The root lattice of E_8 contains 128 vectors of the form $(\pm\frac{1}{2}, \ldots, \pm\frac{1}{2}) \in \mathbb{R}^8$ satisfying the constraint $\sum_{i=1}^{8} x_i \in 2\mathbb{Z}$. This constraint has a structural parallel with the involutive nature of the generative operator: the relation (P3) implies two-fold periodicity in the dynamics, which at the level of discrete state labels selects configurations with an even number of sign changes — precisely the parity condition. This is recorded as a *structural observation*, not a derivation: no embedding of the modal algebra into $\mathbb{R}^8$ has been constructed from the postulates, and the identification of the states with binary vectors has not been derived. The parallel may guide the construction of Steps C–E.

9 Testable Predictions

Falsification conditions — the framework is ruled out if:

1. WEP experiments find correlations with binding energy at $\eta < 10^{-16}$ inconsistent with the synchronisation mechanism.
2. The neutron-star central-density window is shown to be $(2\sqrt{2}-1)\rho_0$ for reasons unrelated to closure structure, or observations rule out the window entirely.
3. The E_8 chain (Steps C–E) cannot be completed, establishing $I(4)^2 = 240$ as a numerical coincidence.
4. The proton charge radius stabilises outside $4\bar{\lambda}_p \pm 0.002$ fm.
5. G_{eff}/G is measured to exceed $\sqrt{16/15} \approx 1.033$ at any density — this is the hard ceiling of the corrected formula and cannot be exceeded within the framework.

Current observational tension. PSR J0952-0607 has a reported mass of $2.35 \pm 0.17\, M_\odot$ [18]. The corrected framework predicts $M_{\max} = 2.037\, M_\odot$ with the SLy EoS, which lies 1.8σ below this measurement. The tension is not yet significant at the 2σ level and depends on the equation of state assumed. A stiffer EoS (e.g. MS1) could accommodate higher masses within the framework; this comparison requires solving TOV with multiple EoS, which is identified as a priority for future work. If $M_{\max} > 2.1\, M_\odot$ is confirmed at $> 3\sigma$ for any object, the ceiling $\sqrt{16/15}$ would need revision.

9.1 WEP

[DERIVED] $\eta = 0$ for ordinary matter (LCM synchronisation, Section 4). No deviation predicted for Ti vs. Pt in MICROSCOPE. Deviations possible only in phase-transition regimes.

9.2 Neutron-star density window — main new prediction

Prediction (parameter-free): Stable neutron stars occupy central densities $\rho_{\text{onset}} \leq \rho_c \leq \rho_{\text{collapse}}$, where $\rho_{\text{onset}} \approx 1.52\,\rho_0$ and $\rho_{\text{collapse}} \approx 4.30\,\rho_0$, with window width $(2\sqrt{2}-1)\rho_0 \approx 1.83\,\rho_0$ exactly.

Testable consequence: No stable neutron star should have $\rho_c < 1.52\,\rho_0$ (would not have completed the $n = 6$ transition) or $\rho_c > 4.30\,\rho_0$ (would have collapsed to black hole via unstable $n = 7$). This constrains the equation of state independently of nuclear physics models.

Current observational status. The density window 1.52–4.30 ρ_0 for stable neutron stars is consistent with existing nuclear-theory equations of state [19] and with the ∼1.8–2.4 $M_\odot$ mass range of observed neutron stars. A precise comparison requires the TOV solution (open problem).

9.3 Maximum mass

[DERIVED] Corrected result (v10.0). TOV with $G_{\text{eff}}(\rho) = G \cdot \omega_m(n(\rho))/\omega_m(4)$ yields $M_{\max} = 2.037\, M_\odot$ (SLy EoS), compatible with PSR J0740+6620 ($2.08 \pm 0.07\, M_\odot$). The coupling ceiling $G_{\text{eff}}/G \leq \sqrt{16/15} \approx 1.033$ gives a parameter-free deviation $\Delta M_{\max} \approx -0.03\, M_\odot$ from GR — a falsifiable prediction. The density window $(2\sqrt{2}-1)\rho_0$ is unaffected.

10 Discussion

10.1 Comparison with emergent-gravity frameworks

Table 11: Comparison with emergent-gravity approaches.

Approach	$1/r^2$	G	Algebra	WEP
Newton/Einstein	Assumed	Fundamental	—	Postulate
Verlinde	Assumed	Proportional	—	Not addressed
Jacobson	Assumed	Via $S \propto A/G$	—	Not addressed
This work	Derived	$\alpha(R_V^2/m_p)a_m$	$\mathfrak{so}(4)$	Emergent

10.2 Version history and reviewer responses

An external reviewer correctly identified three weaknesses in v9.2: (i) calibration circularity in α; (ii) WEP prediction numerically falsified; (iii) lack of a sharp quantitative prediction.

Versions 9.3–9.5 addressed (ii) and (iii): the WEP prediction was replaced by the LCM synchronisation argument ($\eta = 0$ for ordinary matter); the density window $(2\sqrt{2} - 1)\rho_0$ provided a parameter-free prediction. The calibration circularity (i) persists and is acknowledged.

Version 9.6 solved the TOV numerically with $G_{\text{eff}} = G \cdot I(n)/I(4)$, obtaining $M_{\max} = 0.70\, M_\odot$ — incompatible with observations. This falsification was reported honestly, and the direction of correction was identified: the formula treated all modal degrees of freedom equally, ignoring the suppression of the V-mode channel.

Version 10.0 implements the correction. The corrected formula $G_{\text{eff}} = G \cdot \omega_m(n)/\omega_m(4)$ is derived from the existing postulates with no free parameters. TOV integration gives $M_{\max} = 2.037\, M_\odot$ — compatible with all current observations. The hard ceiling $G_{\text{eff}}/G \leq \sqrt{16/15}$ is a new structural prediction of the framework.

10.3 Program notes

[SPECULATIVE] Cosmological prediction. The covariant extension yields $d_L(z; q)$ with constant deceleration parameter q. For $q = -0.55$, deviations from ΛCDM reach $\sim$ 1–3% at $z \sim$ 1–2, testable by LSST, Euclid, and DESI. The value $q = -0.55$ is phenomenological; its derivation from the relational framework is an open problem. This prediction is recorded as a program note, not a central result.

[SPECULATIVE] Propagation invariant and limiting speed. The exact relation $R_V(n)\,\omega_V(n) = c$ (Remark 2) holds at every closure level without assuming special relativity. **[SPECULATIVE]** The interpretation of c as a limiting speed in the relational framework requires showing that no signal can propagate faster than $R_V\omega_V$ — this is a structural analogy, not a derivation. Establishing this connection rigorously is recorded as a program note.

[SPECULATIVE] Orbital resonances. The LCM synchronisation mechanism underlying the WEP has structural parallels with orbital resonances in planetary rings [20]. The Cassini Division (2:1 resonance with Mimas, lcm = 2) corresponds to a small-LCM state where perturbations accumulate; stable ring regions correspond to large-LCM states. This suggests the relational principle may apply from 10^{-15} m (hadronic) to 10^{8} m (planetary rings). No quantitative mapping from orbital frequencies to closure parameters has been developed; this remains a qualitative structural analogy.

10.4 Limitations and future priorities

Current limitations: α calibrated (one empirical input); isotropy not derived; E_8 Steps C–E pending; identification $R_V(n) = d(\rho)/2$ not derived from postulates; prime-instability argument is structural motivation.

Priority future work:

1. Solve TOV with multiple EoS (APR4, MS1, WFF1) to map the $M_{\max}$ prediction as a function of stiffness and compare with PSR J0952-0607 ($2.35 \pm 0.17\, M_\odot$).
2. Derive $R_V(n) = d(\rho)/2$ from the postulates — closes the stellar sector.
3. Derive the channel-rate scaling $G_{\text{eff}} \propto \omega_V^2 I(n)$ from Postulates 1–4 directly.
4. Determine n_c algebraically — closes α_G.
5. Complete E_8 chain Steps C–E.
6. Derive isotropy; prove prime-instability criterion.

11 Conclusion

Established (no free parameters):

- $\times$ is the quaternionic cross product; $\sigma = \pm 1$ is left/right chirality.
- $\{V, m, a\} \to \mathfrak{so}(3)$; both orientations $\to \mathfrak{so}(4)$.
- $I(4)^2 = 240 = \#\text{roots}(E_8)$.
- $\eta(n) = \alpha I(n)$; $\alpha_G = I(4)/I(n_c)$.
- $2R_V(4) = R_V(6)$ exactly.
- $R_V(n) = c\,\tau_V(n)$ — causal coherence length (Remark 2).
- Density window $(2\sqrt{2} - 1)\rho_0$ for neutron-star stability.
- WEP from LCM synchronisation: $\eta = 0$ for ordinary matter (under coupling hypothesis).
- Propagation invariant $R_V(n)\,\omega_V(n) = c$ exactly for all n (Remark 2).
- Corrected $G_{\text{eff}} = G \cdot \omega_m(n)/\omega_m(4)$; ceiling $G_{\text{eff}}/G \leq \sqrt{16/15} \approx 1.033$ (under channel-rate hypothesis Eq. 14; the algebra $\omega_V^2 I(n) = \omega_m \omega_N$ is exact; the quadratic scaling $\Gamma \propto \omega_V^2$ is motivated by the involutive structure of (P3) but remains a hypothesis); TOV result $M_{\max} = 2.037\,M_\odot$ (SLy EoS) — compatible.

Established (one calibration):

- $G = \alpha(R_V^2/m_p)a_m$; natural scale $G/\alpha = \sqrt{240}\,\hbar c/m_p^2$.
- Inverse-square law from conservation and isotropy.

Open (in order of priority):

1. Solve TOV with multiple EoS (APR4, MS1, WFF1) to map $M_{\max}$ vs. stiffness and address PSR J0952-0607 ($2.35 \pm 0.17\,M_\odot$).
2. Derive the two-horizon counting $d = 2R_V(n)$ from the algebraic postulates — closes the stellar sector (causal scale $R_V = c\,\tau_V$ already established).
3. Determine n_c algebraically — closes α_G.
4. Complete E_8 chain Steps C–E.
5. Derive isotropy; prove prime-instability criterion formally.

The correction is not an ad hoc fit. It identifies which modal channel carries the gravitational interaction and derives the coupling from mode frequencies. The hard ceiling $G_{\text{eff}}/G \leq \sqrt{16/15} \approx 1.033$ is a non-trivial prediction: the framework forbids large deviations from Newtonian gravity at any density. The density window $(2\sqrt{2} - 1)\rho_0$ and the connection $I(4)^2 = 240 = \#\text{roots}(E_8)$ are unaffected.

A Contraction of $P_{\mu\nu}$

$P_{\mu\nu}P^{\mu\nu} = -\frac{1}{2}\nabla^\mu\psi\nabla_\mu\psi$. $\square$

B Geometric relation $R_V^2 = I(n)R_N R_m$

$I(n)R_N R_m = 2^{n/2}\sqrt{2^n - 1} \cdot \bar{\lambda}_p \cdot \bar{\lambda}_p / \sqrt{(2^n - 1)/2^n} = 2^n \bar{\lambda}_p^2 = R_V^2$. $\square$

C Critical densities: derivation

$\rho_{\text{crit}}(n) = m_p/R_V(n)^3 = m_p/(2^{3n/2}\bar{\lambda}_p^3)$.

$$\rho_{\text{onset}} = \frac{m_p}{(8\bar{\lambda}_p)^3} = \frac{m_p}{512\bar{\lambda}_p^3} \approx 1.52\,\rho_0.$$

$$\rho_{\text{collapse}} = \frac{m_p}{(4\sqrt{2}\bar{\lambda}_p)^3} = \frac{m_p}{128\sqrt{2}\bar{\lambda}_p^3} \approx 4.30\,\rho_0.$$

Ratio: $\rho_{\text{collapse}}/\rho_{\text{onset}} = 512/(128\sqrt{2}) = 4/\sqrt{2} = 2\sqrt{2}$. $\square$

D Parameter accounting

Parameter	Value	Origin	Sensitivity	Status
n	4	Motivated	Discrete	Open: derive
α	3.81×10^{-40}	1 calibration	$\propto G$	1 free param.
$m_p, \hbar, c$	CODATA 2018	Empirical	—	Input
n_c	≈ 131	Estimated	Log	Central open
G_{eff} profile	$G \cdot \omega_m(n)/\omega_m(4)$	Derived	$\leq 3.3\%$	TOV: $M_{\max} = 2.04\,M_\odot$ ✓

E Glossary

Symbol	Definition
$\hat{\aleph}$	Normalised reference
$\sigma \in \{+1, -1\}$	Left/right $\mathbb{H}$-action
$\{U, V, m, a\}$	Four modes
n	Closure depth ($n = 4$: proton; $n = 6$: neutron star)
$I(n) = 2^{n/2}\sqrt{2^n - 1}$	Modal invariant
$\alpha \approx 3.81 \times 10^{-40}$	Relational coupling (calibrated)
$\alpha_G = Gm_p^2/(\hbar c)$	Gravitational coupling
$\eta(n) = \alpha I(n)$	Instability parameter
n_c	Planck depth ($\eta(n_c) = 1$); open
ρ_{onset}	$1.52\,\rho_0$ — start of $n = 6$ collective closure
ρ_{collapse}	$4.30\,\rho_0$ — $n = 7$ instability threshold
$\Delta\rho = (2\sqrt{2} - 1)\rho_0$	Neutron-star stability window
$G_{\text{eff}}(\rho)$	$G \cdot \omega_m(n(\rho))/\omega_m(4)$; ceiling $\sqrt{16/15}G$
$\omega_V(n)$	$\omega_N \cdot 2^{-n/2}$ (visible-mode frequency)
$\omega_m(n)$	$\omega_N \cdot 2^{-n/2}\sqrt{2^n - 1}$ (confinement-mode frequency)
$R_V(n)\,\omega_V(n)$	$= c$ exactly, all n (propagation invariant, Remark 2)

Adversarial AI Cross-Validation (AACV)

Adversarial review using Google Gemini (anonymous session) raised five critiques, all addressed in the paper: (1) circularity in $\eta(n)$ — resolved in §3.2; (2) $\mathfrak{so}(4)$ not novel — novelty is the physical mapping, not the algebra; (3) $n = 4$ post-hoc — acknowledged **[OPEN]** throughout; (4) WEP no numerical β — replaced by LCM argument (§4); (5) $I(4)^2 = 240$ numerology — algebraically exact, pending E_8 Steps C–E.

LLM Disclosure Statement

Models: Anthropic Claude (primary), Google Gemini (AACV), OpenAI ChatGPT (early versions).

Contributions: Matrix commutator verification (Theorems 1–2); dimensional consistency; E_8 connection identification; WEP calculation and correction (falsified linear prediction, replacement by LCM argument); neutron-star density-window derivation; critical review at each version.

Human oversight: All claims independently verified by the author. Matrices verified by hand. LLM suggestions that overclaimed (e.g., "exact derivation of α_G", "WEP violations in Ti/Pt") were corrected to current labelling.

Version history: v1.0 (April 2025) → v9.6 → v10.0 → **v10.1** (April 2026). Key milestones: v8.0 algebraic classification; v9.3 WEP correction + stellar hierarchy; v9.5 G_{eff} formula; v9.6 TOV + identification of $G \cdot I(n)/I(4)$ error ($M_{\max} = 0.70\,M_\odot$, incompatible);

v10.0 corrected $G_{\text{eff}} = G \cdot \omega_m(n)/\omega_m(4)$, $M_{\max} = 2.037\, M_\odot$ (compatible); v10.1 $G_{\text{eff}} \propto \omega_V^2 I(n)$ derived from (P3); $R_V \omega_V = c$ proved; $d = 2R_V$ grounded in causal coherence $R_V = c\,\tau_V$ (two-horizon counting remains open).

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